

Theory of Computation

FINAL EXAM

SI	Topic Name	Marks
Group A		
01	Formal Definition of CFG & CNF	2
02	Construct a CFG for given language or regular expression	3
03	Show that grammar is ambiguous by two i. parse trees ii. rightmost or leftmost derivation.	5
04	Convert the following CFG into an equivalent CFG in Chomsky Normal Form	4
05	Using the Pumping lemma show given language are not context free	3
06		
Group B		
08	All about PDA	2
09	Convert following CFGs to an equivalent pushdown automation	4
10	Construct PDA that accepts following language	4
11	Given the implementation-level description of turning machine that accept the following languages	6
12	Give the sequence of configuration for Turing Machine	4
13	Theoretical proof from Turing machine	4
14	Prove Decidability and Undecidability	5
15	What is NP-complete and NP-Hard problems? How can you show that a problem is NP-complete?	3

SEG 4

Definition:

1. Formal Definition of CNF

Construct a CFG that generate the following languages over alphabet Σ is $\{0,1\}$:

- a. $\{w \mid w \text{ contains at least three } 1s\}$
- b. $\{w \mid w \text{ starts and ends with the same symbol}\}$
- c. $\{w \mid \text{the length of } w \text{ is odd}\}$
- d. $\{w \mid \text{the length of } w \text{ is odd and its middle symbol is a } 0\}$
- e. $\{w \mid w = w^R, \text{ that is, } w \text{ is a palindrome}\}$
- f. The empty set
- g. $\{w \mid \text{the length of } w \text{ is odd and its middle symbol is a } 1\}$
- h. $(0 + 1)^*011^*$
- i. $(0+1)^*01^*$
- j. $0^*1(0+1)^*$
- k. $A = \{a^i b^j c^k \mid i=j \text{ or } j=k \text{ where } i, j, k \geq 0\}$.

Show that grammar is ambiguous by two parse tree & rightmost or leftmost derivation.

- a) $S \Rightarrow E, E \Rightarrow E + E \mid E - E \mid E^*E, E \Rightarrow (E) \mid a$ for string " $a+a^*(a-a)$ "
- b) $S \Rightarrow SAB \mid \varepsilon, A \Rightarrow AaB \mid a, B \Rightarrow AS \mid b$ for string $aaaabaabbabab$
- c) $S \Rightarrow iCtS \mid iCtSeS \mid a, C \Rightarrow b$ for string $ibtibttaea$

CNF

Convert the following CFG into an equivalent CFG in Chomsky Normal Form

- | | | |
|---|---|---|
| a) $S \rightarrow ASA \mid aB$
$A \rightarrow B \mid S$
$B \rightarrow b \mid \varepsilon$ | d) $S \rightarrow ASB$
$A \rightarrow AS \mid a \mid \varepsilon$
$B \rightarrow SbS \mid A \mid b$ | g) $S \rightarrow aXbX$
$X \rightarrow aY \mid bY \mid \varepsilon$
$Y \rightarrow X \mid c$ |
| b) $S \rightarrow ASB$
$A \rightarrow aAS \mid a \mid \varepsilon$
$B \rightarrow SbS \mid A \mid bb$ | e) $R \rightarrow aSa \mid bRb \mid s$
$S \rightarrow aTb \mid bTa \mid aS$
$T \rightarrow XTX \mid X \mid \varepsilon$
$X \rightarrow a \mid b$ | h) $R \rightarrow aRb \mid bRb \mid S$
$S \rightarrow aTa \mid bTa$
$T \rightarrow XTX \mid X \mid \varepsilon$
$X \rightarrow a \mid b$ |
| c) $S \rightarrow TX$
$T \rightarrow 0T0 \mid 1T1 \mid \#X$
$X \rightarrow 0X \mid 1X \mid \varepsilon$ | f) $S \rightarrow aSb \mid bY \mid Ya$
$Y \rightarrow bY \mid aY \mid c \mid \varepsilon$ | i) $A \rightarrow BAB \mid B \mid \varepsilon$
$B \rightarrow 00 \mid \varepsilon$ |

Using the Pumping lemma show given language are context free or not:

- 1. $L1 = \{w \mid w\}$

SEG 5

Definition:

1. Formally defined the push down automata. Illustrate the component of PDA with figure.
2. Why do you think push down automata are more powerful than finite automata?
3. What is push down automata?
4. Explain the role of stack in PDA and why is it necessary for some languages?
5. Write down, the difference between finite automata and push down automata?

Construct PDA that accepts following language

- a. $\{0^n 1^n | n \geq 0\}$
- b. $L = \{a^i b^j c^k | i, j, k \geq 0 \text{ and } i+j=k\}$
- c. $L = \{t^i s^j a^k | i, j, k \geq 0 \text{ and } i=j \text{ or } j=k\}$
- d. $L = \{a^{2n} b^n c^n | n \geq 0\}$
- e. $L = \{a^n b^{2n} | n \geq 0\}$
- f. Construct PDA that recognizes any arithmetic expression involving +, * and any one digit integer.

Convert following CFGs to an equivalent pushdown automation

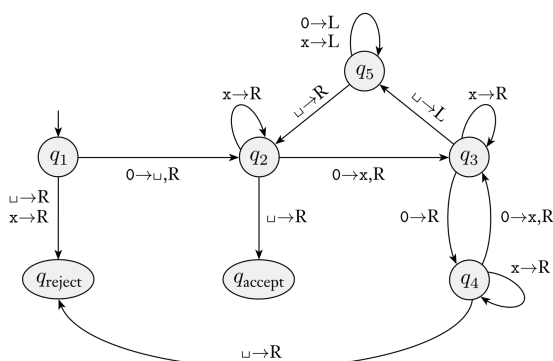
- a.

SEG 06

Given the implementation-level description of turning machine that accept the following languages

- a. $L = \{w | w \text{ has an equal number of } 0\text{'s and } 1\text{'s}\}$
- b. $L = \{ww^R | w \text{ is any string of } 0\text{'s and one } 1\text{'s}\}$
- c. $L = \{a^n b^n a^n | n \geq 0\}$
- d. $L = \{w | w \text{ is it string with } 0\text{'s and } 1\text{'s contains } 1\text{'s in a multiple of } 3\}$
- e. $L = \{w | w \text{ does not contain twice as many } 0\text{'s as } 1\text{'s}\}$
- f. $L = \{a^{2n} b^n c^n | n \geq 0\}$

Give the sequence of configuration that the machine enters started with the following string



- i. 00
- ii. 0000
- iii. 0000000

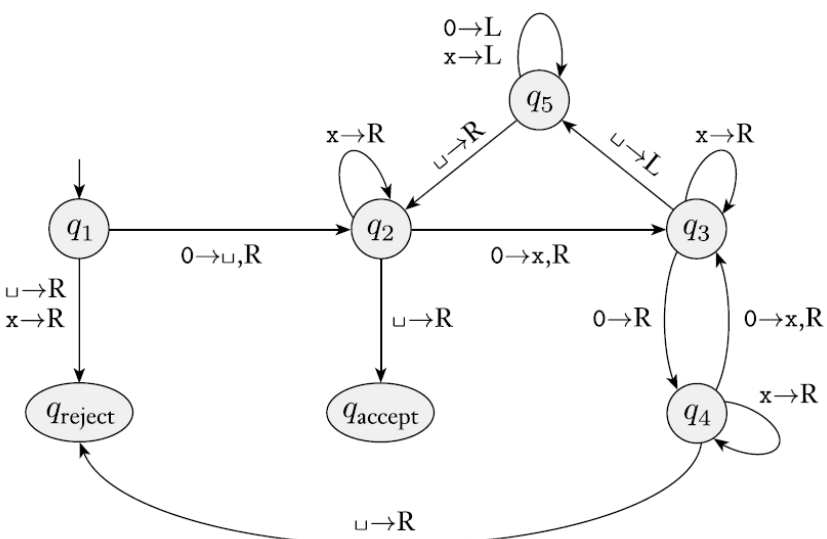
SEG 07

1. What is the Church Turing Thesis?
2. Define decidability and undecidability in context of theory of computing.
3. Find out whether the following problem is decidable or not.
 - i) Is a number 'm' prime
 - ii) Determining if a given graph G is connected?
4. Consider the language
$$A_{DFA} = \{(B, w) \mid B \text{ is a DFA that accepts input string } w\}$$
Prove that, " A_{DFA} is a decidable language"
5. Let the acceptance language of NFA
$$A_{NFA} = \{(N, w) \mid N \text{ is an NFA that accepts input string } w\}.$$
Prove that A_{NFA} is decidable.
6. Introduce the diagonalization method/argument.
7. Produce an illustrative example/application of it to show that the set of real numbers is uncountable
8. What are the key ideas to formally prove that the non-deterministic TM has an equivalent deterministic TM?
9. Let a language L is define as
$$L = \{(M, w) \mid M \text{ is a TM and } M \text{ halts on input } w\}.$$
Show that L is undecidable.
10. Prove that the Halting Problem for Turing Machine is undecidable
11. Show that the set of infinitely long binary sequences is uncountable.

SEG 08

1. What is polynomial-time reducibility?
2. What do you understand by polynomial and exponential time complexity?
3. What are NP complete and NP-hard problems?
4. How you can show that a problem is NP-complete?
5. Define the complexity classes P, NP, NP-hard and NP-complete?

	Group-A			
			CO	DL
1.				
a)	Describe the components of context-free grammar. OR Define parse tree and derivation.	3	CO1	U
b)	What is ambiguity? Determine whether this grammar is ambiguous. $S \rightarrow iEtS \mid iEtSeS \mid a$ $E \rightarrow b$	3	CO2	N
c)	Consider the following grammar $S \rightarrow aSb \mid bY \mid Ya$ $Y \rightarrow bY \mid aY \mid \epsilon$ Show how to derive the string abbab using this grammar using a left-most derivation. Draw the parse tree for the string. OR Show how to derive the string abbab using this grammar using a right-most derivation. Draw the parse tree for the string.	4	CO2	A
2.				
a)	How do you use the pumping lemma to determine if a language is context-free? OR Can you describe a regular language using context-free grammar? Illustrate using an example.	3	CO1	N
b)	Can you give a context-free grammar (CFG) for the following languages over the alphabet $\Sigma = \{a,b\}$:- All strings in the language L: $\{a^n b^{2n} c^{4n} \mid n \geq 0\}$ If you cannot, justify the reason.	3	CO2	E
c)	Give a context-free grammar (CFG) for each of the following languages over the alphabet $\Sigma = \{a,b\}$:- - All strings in the language L: $\{a^n b^{2n} \mid n \geq 0\}$ - All nonempty strings that read the same from left or right.	4	CO2	C

	Group-B			
3.				
a)	What are ϵ -rule and unit rules?	2	C02	U
b)	Construct a pushdown automaton that recognizes the following language L: $\{a^n b^{2n} \mid n \geq 0\}$ OR Construct a pushdown automaton that recognizes all even nonempty palindromes.	4	C03	C
c)	Convert any one of the following context-free grammar (CFG) to an equivalent pushdown automaton $S \rightarrow aSb \mid bY \mid Ya$ $Y \rightarrow bY \mid aY \mid c \mid \epsilon$ OR $S \rightarrow aXbX$ $X \rightarrow aY \mid bY \mid \epsilon$ $Y \rightarrow X \mid c$	4	C01	N
4.				
a)	Consider the following Turing machine.  Give the sequence of configurations that the machine enters when started with the following strings. 0000 000000	4	C03	A

b)	Give the implementation-level description of a Turing machine that decides the following languages (any two) i. $L: \{a^n b^{2n} c^{4n} \mid n \geq 0\}$ ii. $\{w \mid w \text{ is a non-empty odd palindrome}\}$	6	C03	C
5.				
a)	Convert any one of the following CFG into an equivalent CFG in Chomsky normal form. $S \rightarrow aSb \mid bY \mid Ya$ $Y \rightarrow bY \mid aY \mid c \mid \epsilon$ OR $S \rightarrow aXbX$ $X \rightarrow aY \mid bY \mid \epsilon$ $Y \rightarrow X \mid c$	4	C02	A
b)	Find out whether the following problem is decidable or not: Is a number 'm' prime?	3	C04	E
c)	What are NP-complete and NP-Hard problems? How can you show that a problem is NP-complete?	3	C05	U

	Group-A			
		CO	DL	
1.				
a)	Describe the components of context-free grammar. Is any of the components have a similarity with any of the components of regular language or finite automata? OR Define and differentiate between the following. a. Derivation and Parse Tree b. Leftmost and Rightmost derivation	CO1	U	3
b)	What is ambiguity? Determine whether the following grammar is ambiguous. $S \rightarrow AB$ $A \rightarrow aA / abA / \epsilon$ $B \rightarrow bB / abB / \epsilon$	CO2	N	3
c)	Consider the following grammar $S \rightarrow SSx / SSy / SSz / a / b / c$ Show how to derive the string cabxycz using this grammar using a left-most derivation. Draw the parse tree for the string. OR Show how to derive the string cabxycz using this grammar using a right-most derivation. Draw the parse tree for the string.	CO2	A	4
2.				
a)	Show using the pumping lemma which of the following languages are context-free. i. $L1 = \{w / w \in a^n b^n c^{2n} / n \geq 0\}$ ii. $L2 = \{w / w \in a^n b^n c^n / n \geq 0\}$ OR Describe the following languages using a context-free grammar. a. 0^*1^* b. $1(01)^*$ c. $(11 \cup 0)^*$	CO2	N	3
b)	Give a context-free grammar (CFG) for each of the following languages over the alphabet $\Sigma = \{a,b\}$:- i. $L = \{a^{2n}b^m c^n / n \geq 0\}$ ii. $L = \{a^m b^n c^{m+n} / n \geq 0\}$	CO2	C	4

c)	Convert the following CFG into an equivalent CFG in Chomsky normal form. $R \rightarrow aSa \mid bRb \mid S$ $S \rightarrow aTb \mid bTa \mid aS$ $T \rightarrow XTX \mid X \mid \varepsilon$ $X \rightarrow a \mid b$	C02	N	3
	Group-B			
3.				
a)	How ε -rules are removed when converting a grammar to Chomsky normal form? OR Why do you think pushdown automata are more powerful than finite automata?	C02	U	2
b)	Construct a pushdown automaton that recognizes the following language $L = \{a^{2n}b^m c^n \mid n \geq 0\}$ OR Construct a pushdown automaton that recognizes any arithmetic expression involving +, *, and any one-digit integer.	C03	C	4
c)	Convert any one of the following context-free grammar (CFG) to an equivalent pushdown automaton $S \rightarrow XYm \mid XYn$ $X \rightarrow aX \mid \varepsilon$ $Y \rightarrow bY \mid \varepsilon$ OR $S \rightarrow aXbY$ $X \rightarrow aYa \mid \varepsilon$ $Y \rightarrow bXb \mid c \mid \varepsilon$	C03	N	4
4.				
a)	What is the Church-Turing thesis?	C04	U	2
b)	Can you run a nondeterministic algorithm on a deterministic machine instead of a nondeterministic one? If your answer is yes, then explain how you can do it and how the running time will be affected. If your answer is no, then explain why it will not be possible.	C05	E	4
c)	Give the implementation-level description of a Turing machine that decides the following languages: $L = \{a^{2n}b^n c^n \mid n \geq 0\}$.	C03	A	4

5.				
a)	Let the language $A_{DFA} = \{(B, w) \mid B \text{ is a DFA that accepts input string } w\}$. Prove that " A_{DFA} is a decidable language."	C04	E	3
b)	Show that the set of infinitely long binary sequences is uncountable.	C04	N	3
c)	What are N Let, $A_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM and } M \text{ accepts } w \}$ and $HALT_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM and } M \text{ halts on input } w \}$ Prove that if A_{TM} is undecidable then $HALT_{TM}$ is also undecidable.	C05	N	4

Syllabus

CFG

PDA

TM

Decidability, Reduction

Complexity – Dropped

Question Pattern followed

1. Definition: CFG, parse tree, derivation
2. Ambiguity and determining if a particular grammar is ambiguous
3. Construct derivation and parse tree
4. Simple example with pumping lemma (with or), the relationship between CFL and regular language
5. Construct CFG from the description of a language
6. Chomsky normal form and converting a grammar to Chomsky normal form
7. Definition and basic knowledge type question related to CFG, CFL, and PDA
8. Construct PDA from the description of a language
9. Convert CFG to an equivalent PDA
10. Simulate a given Turing machine – Dropped
11. Definition and basic knowledge type question related to the Turing machine
12. Theoretical proof from Turing machine
13. Construct a Turing machine from the description of a language
14. Decidability and proving a simple problem as decidable
15. Simple proof related to the decidable problems
16. Definition and simple example related to complexity theory – Dropped
17. Simple problem and proof related to the reduction